

VALIDATION PLAN
OPERATIONAL QUALIFICATION
CLINICAL MODULE

JR Validated Environment — Clinical Module v2.0
Module 1: Study Design | Module 2: Diagnostic Accuracy

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Reviewer	[Name]
Approver	[Name]

Version History

Version	Date	Author	Description
1.0	2026-07-15	Joep Rous	Initial release. Covers the 3 clinical study design scripts (Module 1); 36 automated test cases, UR-CLIN-001 to UR-CLIN-036.
2.0	2026-07-16	Joep Rous	Scope extended to Module 2 — Diagnostic Accuracy: jrc_clinical_dx_accuracy, jrc_clinical_dx_ss and jrc_clinical_dx_roc add 51 test cases (UR-CLIN-037 to UR-CLIN-087), for 87 in total. UR-CLIN-001 to UR-CLIN-036 and their test cases are unchanged from Rev 1 and retain their numbering.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-07-16
Reviewer	[Name]		
Approver (QA)	[Name]		

Purpose and Scope

This Operational Qualification (OQ) Validation Plan defines the test cases for the Clinical module of the JR Validated Environment. It specifies the user requirements, test cases, acceptance criteria, and traceability needed to demonstrate that each clinical script performs correctly for its intended use under 21 CFR 820.70(i) and ISO 13485, in support of clinical investigations planned under ISO 14155 / ICH E9 principles.

This plan covers 87 automated test cases executed using the pytest framework via the admin_clinical_oq runner. Test evidence is written to `~/jrscrip<PROJECT_ID>/validation/clinical_oq_execution_<timestamp>.txt`.

Module 1 — Clinical Study Design (Rev 1). Three scripts compute two-arm parallel-group sample sizes: `jrc_clinical_ss_means` (continuous endpoint), `jrc_clinical_ss_props` (binary endpoint, risk difference), and `jrc_clinical_ss_survival` (time-to-event, log-rank). Each supports the applicable hypothesis frameworks (superiority, non-inferiority, and — except survival — equivalence/TOST) via the `--framework` flag, with allocation ratio, dropout inflation, and sensitivity scenario tables computed in the validated R layer. Statistical methods: Chow, Shao & Wang (2008), Sample Size Calculations in Clinical Research, 2nd ed., Chapters 3 and 4 (normal approximations); Schoenfeld (1983) events formula for the log-rank test. All three scripts use base R only.

Module 2 — Diagnostic Accuracy (added at Rev 2). Three scripts evaluate a diagnostic test against a binary reference standard. `jrc_clinical_dx_ss` sizes a diagnostic study by target CI half-width or against a performance goal, sizing the reference-positive and reference-negative groups separately and converting to total enrolment via the prevalence (Buderer 1996). `jrc_clinical_dx_accuracy` reports the 2x2 operating characteristics — sensitivity, specificity, PPV, NPV, LR+, LR- and accuracy — with Clopper-Pearson exact or Wilson intervals, likelihood-ratio intervals on the log scale (Simel 1991), and Bayes-adjusted PPV/NPV at a stated prevalence per FDA (2007), Statistical Guidance on Reporting Results from Studies Evaluating Diagnostic Tests. `jrc_clinical_dx_roc` reports the empirical ROC curve, the AUC with a DeLong (1988) confidence interval, and the Youden-optimal cutoff. Unlike Module 1, `jrc_clinical_dx_accuracy` and `jrc_clinical_dx_roc` consume a data file.

Statistical implementation note. The AUC and its DeLong variance are implemented in base R using the midrank algorithm of Sun & Xu (2014), rather than delegated to an external package; no ROC package is pinned in this environment. Two independent controls apply. First, the acceptance criterion for the AUC (TC-CLIN-DXROC-003) is taken from the original literature — Hanley & McNeil (1982), Radiology 143:29-36, Table 1, which reports an area of 0.893 — rather than from this implementation's own output. Second, `jrc_clinical_dx_roc` recomputes the AUC at run time by the definitional $O(m \cdot n)$ Mann-Whitney double sum and halts if the two disagree, so every execution re-verifies the algorithm against its definition. `jrc_clinical_dx_roc` imports ggplot2 to draw its plot; the statistics are base R. The remaining five clinical scripts use base R only.

User Requirements

The following user requirements are tested by this plan.

UR ID	Requirement
UR-CLIN-001	<code>jrc_clinical_ss_means</code> shall accept valid superiority inputs and exit 0 with the report header present.
UR-CLIN-002	<code>jrc_clinical_ss_means</code> shall compute the superiority per-arm and total sample size per the Chow ch. 3 normal-approximation formula.
UR-CLIN-003	<code>jrc_clinical_ss_means</code> shall report the significance level, sidedness, and the resulting z-value.

UR-CLIN-004	jrc_clinical_ss_means shall compute the non-inferiority sample size per the Chow ch. 3 margin formula at one-sided alpha.
UR-CLIN-005	jrc_clinical_ss_means shall inflate the evaluable n to an enrolled N as $n/(1-\text{dropout})$, rounded up per arm.
UR-CLIN-006	jrc_clinical_ss_means shall compute the equivalence sample size with the TOST beta term $z(1-(1-\text{power})/2)$.
UR-CLIN-007	jrc_clinical_ss_means shall support unequal allocation via --ratio (treatment:control).
UR-CLIN-008	jrc_clinical_ss_means shall print an n-vs-SD scenario table over $SD \times 0.8 \dots 1.2$ when --sensitivity is given.
UR-CLIN-009	jrc_clinical_ss_means shall exit non-zero when superiority is requested without a meaningful difference.
UR-CLIN-010	jrc_clinical_ss_means shall exit non-zero when non-inferiority is requested without a margin.
UR-CLIN-011	jrc_clinical_ss_means shall exit non-zero when the assumed true difference is not inside the equivalence zone.
UR-CLIN-012	jrc_clinical_ss_means shall exit non-zero when --power is not strictly between 0 and 1.
UR-CLIN-013	jrc_clinical_ss_means shall exit non-zero on any unrecognized argument.
UR-CLIN-014	jrc_clinical_ss_means shall exit non-zero and report RENV_PATHS_ROOT when called directly via Rscript without the wrapper.
UR-CLIN-015	jrc_clinical_ss_props shall accept valid superiority inputs and exit 0 with the report header present.
UR-CLIN-016	jrc_clinical_ss_props shall compute the superiority sample size per the Chow ch. 4 unpooled risk-difference formula.
UR-CLIN-017	jrc_clinical_ss_props shall compute the NI sample size, defaulting the treatment rate to the control rate when --p-treat is omitted.
UR-CLIN-018	jrc_clinical_ss_props shall compute the equivalence sample size with the TOST beta term.
UR-CLIN-019	jrc_clinical_ss_props shall exit non-zero when superiority is requested without the expected treatment rate.
UR-CLIN-020	jrc_clinical_ss_props shall exit non-zero when p-treat equals p-control under superiority.
UR-CLIN-021	jrc_clinical_ss_props shall exit non-zero when non-inferiority is requested without a margin.
UR-CLIN-022	jrc_clinical_ss_props shall exit non-zero when --p-control is not strictly between 0 and 1.
UR-CLIN-023	jrc_clinical_ss_props shall print an n-vs-p-control scenario table over $p\text{-control} \times 0.8 \dots 1.2$ when --sensitivity is given.
UR-CLIN-024	jrc_clinical_ss_props shall inflate the evaluable n to an enrolled N as $n/(1-\text{dropout})$, rounded up per arm.
UR-CLIN-025	jrc_clinical_ss_props shall exit non-zero and report RENV_PATHS_ROOT when called directly via Rscript without the wrapper.
UR-CLIN-026	jrc_clinical_ss_survival shall accept valid superiority inputs and exit 0 with the report header present.
UR-CLIN-027	jrc_clinical_ss_survival shall compute required events per Schoenfeld and convert to subjects via the event probability.
UR-CLIN-028	jrc_clinical_ss_survival shall compute NI events using $\log(\text{margin}) - \log(\text{true HR})$ at one-sided alpha.
UR-CLIN-029	jrc_clinical_ss_survival shall exit non-zero when --framework equivalence is requested.
UR-CLIN-030	jrc_clinical_ss_survival shall exit non-zero when superiority is requested with a hazard ratio of exactly 1.
UR-CLIN-031	jrc_clinical_ss_survival shall exit non-zero when the NI margin on the hazard ratio is ≤ 1 .
UR-CLIN-032	jrc_clinical_ss_survival shall exit non-zero when the assumed true HR is not less than the margin.
UR-CLIN-033	jrc_clinical_ss_survival shall exit non-zero when the event probability is missing or out of (0, 1].
UR-CLIN-034	jrc_clinical_ss_survival shall print an n-vs-event-prob scenario table over $\text{event-prob} \times 0.8 \dots 1.2$ when --sensitivity is given.
UR-CLIN-035	jrc_clinical_ss_survival shall inflate the evaluable n to an enrolled N as $n/(1-\text{dropout})$, rounded up per arm.
UR-CLIN-036	jrc_clinical_ss_survival shall exit non-zero and report RENV_PATHS_ROOT when called directly via Rscript without the wrapper.
UR-CLIN-037	jrc_clinical_dx_accuracy shall accept a valid id/reference/result CSV and exit 0 with the report header present.
UR-CLIN-038	jrc_clinical_dx_accuracy shall cross-tabulate the index test against the reference standard and report TP, FP, FN, TN with their margins.
UR-CLIN-039	jrc_clinical_dx_accuracy shall compute sensitivity as $TP/(TP+FN)$ with a two-sided Clopper-Pearson exact confidence interval.

UR-CLIN-040	jrc_clinical_dx_accuracy shall compute specificity as $TN/(TN+FP)$ with a two-sided Clopper-Pearson exact confidence interval.
UR-CLIN-041	jrc_clinical_dx_accuracy shall compute overall accuracy as $(TP+TN)/N$ with a two-sided exact confidence interval.
UR-CLIN-042	jrc_clinical_dx_accuracy shall compute LR+ as $sens/(1-spec)$ with the log (Simel 1991) confidence interval.
UR-CLIN-043	jrc_clinical_dx_accuracy shall compute LR- as $(1-sens)/spec$ with the log (Simel 1991) confidence interval.
UR-CLIN-044	jrc_clinical_dx_accuracy shall report PPV and NPV computed from the study's own row totals, state the study prevalence used, and warn that they are valid only if it reflects the intended-use population.
UR-CLIN-045	jrc_clinical_dx_accuracy shall offer a Wilson score interval via <code>--ci wilson</code> , leaving the point estimate unchanged.
UR-CLIN-046	jrc_clinical_dx_accuracy shall, given <code>--prevalence P</code> , additionally report PPV and NPV recomputed by Bayes' theorem at P.
UR-CLIN-047	jrc_clinical_dx_accuracy shall honour <code>--conf</code> , producing a strictly narrower interval at a lower confidence level.
UR-CLIN-048	jrc_clinical_dx_accuracy shall accept arbitrary category labels when the positive label is named with <code>--positive</code> .
UR-CLIN-049	jrc_clinical_dx_accuracy shall drop rows with a missing id, reference or result, and report how many were dropped.
UR-CLIN-050	jrc_clinical_dx_accuracy shall reject an unrecognised <code>--ci</code> method.
UR-CLIN-051	jrc_clinical_dx_accuracy shall reject a <code>--prevalence</code> outside (0, 1).
UR-CLIN-052	jrc_clinical_dx_accuracy shall reject a CSV lacking a required column and name the missing column.
UR-CLIN-053	jrc_clinical_dx_accuracy shall reject a CSV containing duplicate ids.
UR-CLIN-054	jrc_clinical_dx_accuracy shall reject category labels it cannot map to positive/negative, and say so.
UR-CLIN-055	jrc_clinical_dx_accuracy shall refuse to report when the reference standard has no negative subjects.
UR-CLIN-056	jrc_clinical_dx_accuracy shall refuse to run outside the validated environment (RENV_PATHS_ROOT unset).
UR-CLIN-057	jrc_clinical_dx_ss shall accept valid precision-method inputs and exit 0 with the report header present.
UR-CLIN-058	jrc_clinical_dx_ss shall size each reference arm for a target CI half-width and convert to total enrolment via the prevalence, per Buderer (1996).
UR-CLIN-059	jrc_clinical_dx_ss shall size each reference arm to reject a performance goal at the stated power, and convert to total enrolment.
UR-CLIN-060	jrc_clinical_dx_ss shall report which reference arm drives the total enrolment.
UR-CLIN-061	jrc_clinical_dx_ss shall leave the per-arm requirements unchanged when the prevalence changes, while raising the total enrolment.
UR-CLIN-062	jrc_clinical_dx_ss shall, given <code>--sensitivity</code> , print total N across a range of prevalence values with the assumed value marked.
UR-CLIN-063	jrc_clinical_dx_ss shall inflate the evaluable N to an enrolled N for the stated unevaluable fraction.
UR-CLIN-064	jrc_clinical_dx_ss shall default to a two-sided z for the precision method and a one-sided z for the hypothesis method.
UR-CLIN-065	jrc_clinical_dx_ss shall reject the precision method without <code>--halfwidth</code> .
UR-CLIN-066	jrc_clinical_dx_ss shall reject the hypothesis method without <code>--sens-goal</code> .
UR-CLIN-067	jrc_clinical_dx_ss shall reject a performance goal at or above the expected value.
UR-CLIN-068	jrc_clinical_dx_ss shall reject a <code>--prevalence</code> outside (0, 1).
UR-CLIN-069	jrc_clinical_dx_ss shall reject an unrecognised <code>--method</code> .
UR-CLIN-070	jrc_clinical_dx_ss shall refuse to run outside the validated environment (RENV_PATHS_ROOT unset).
UR-CLIN-071	jrc_clinical_dx_roc shall accept a valid id/reference/score CSV and exit 0 with the report header present.
UR-CLIN-072	jrc_clinical_dx_roc shall save a two-panel PNG to <code>~/Downloads/</code> .
UR-CLIN-073	jrc_clinical_dx_roc shall reproduce the area under the ROC curve published by Hanley & McNeil (1982) for their Table 1 rating-scale data.
UR-CLIN-074	jrc_clinical_dx_roc shall report the AUC with its DeLong standard error and a two-sided confidence interval.
UR-CLIN-075	jrc_clinical_dx_roc shall score tied scores as 0.5 in the Mann-Whitney kernel.
UR-CLIN-076	jrc_clinical_dx_roc shall report the cutoff maximising Youden's $J = \text{sensitivity} + \text{specificity} - 1$, with J

	and the operating point.
UR-CLIN-077	jrc_clinical_dx_roc shall report the cutoff as an observed score value together with the explicit rule that applies it.
UR-CLIN-078	jrc_clinical_dx_roc shall test the AUC against 0.5 using the DeLong standard error and report z and p.
UR-CLIN-079	jrc_clinical_dx_roc shall honour --direction lower by reflecting the AUC about 0.5.
UR-CLIN-080	jrc_clinical_dx_roc shall report that no cutoff exists, rather than an infinite threshold, when no candidate cutoff beats chance in the stated direction; and shall warn when AUC < 0.5.
UR-CLIN-081	jrc_clinical_dx_roc shall offer a logit-transformed AUC interval via --ci-method logit, leaving the point estimate unchanged.
UR-CLIN-082	jrc_clinical_dx_roc shall honour --conf, producing a strictly narrower interval at a lower confidence level.
UR-CLIN-083	jrc_clinical_dx_roc shall reject an unrecognised --direction.
UR-CLIN-084	jrc_clinical_dx_roc shall reject a CSV lacking a required column and name the missing column.
UR-CLIN-085	jrc_clinical_dx_roc shall reject a CSV containing duplicate ids.
UR-CLIN-086	jrc_clinical_dx_roc shall refuse to run when the reference standard has fewer than two negative subjects.
UR-CLIN-087	jrc_clinical_dx_roc shall refuse to run outside the validated environment (RENV_PATHS_ROOT unset).

Test Environment

All test cases are executed via the admin_clinical_oq runner using pytest with the shared OQ virtual environment at `~/.venvs/<PROJECT_ID>_oq/`. Evidence is written to `~/.jrscrip<PROJECT_ID>/validation/`.

The four sample-size scripts (jrc_clinical_ss_means, jrc_clinical_ss_props, jrc_clinical_ss_survival, jrc_clinical_dx_ss) are parameter-only: study design happens before data exist, so their test cases pass parameters on the command line and assert on the report. The two analysis scripts added at Rev 2 (jrc_clinical_dx_accuracy, jrc_clinical_dx_roc) consume a CSV; their fixtures are fixed, committed files in `repos/clinical/oq/data/`, so every execution of the suite is reproducible.

Item	Specification
Language	R ($\geq$ 4.3.0 recommended)
R packages	base R only, except jrc_clinical_dx_roc, which imports ggplot2 (pinned in admin/R_requirements.txt) to draw its plot. All statistics in all six scripts are base R.
Test framework	pytest ($\geq$ 8.0)
Test runner	admin_clinical_oq
OQ venv	<code>~/.venvs/<PROJECT_ID>_oq/</code> (shared with admin_oq)
Test data	<code>repos/clinical/oq/data/</code> — fixed committed CSV fixtures (Rev 2; the Rev 1 suite used none)
Evidence output	<code>~/.jrscrip<PROJECT_ID>/validation/clinical_oq_execution_<timestamp>.txt</code>
Validation plan ID	JR-VP-CLIN-001 Rev 2

Test Cases — jrc_clinical_ss_means

TC-CLIN-MEANS-001 Superiority happy path — exit 0

Tests UR(s)	UR-CLIN-001
Command / action	<code>jrc_clinical_ss_means --framework superiority --power 0.80 --alpha 0.05 --sides 2 --sd 10 --delta 5</code>
Rationale	Confirms the script accepts standard valid inputs and completes successfully.
Pass criterion	Exit code = 0. Output contains 'Clinical sample size' and 'continuous endpoint'.

TC-CLIN-MEANS-002 Superiority numeric correctness

Tests UR(s)	UR-CLIN-002
Command / action	<code>jrc_clinical_ss_means --framework superiority --power 0.80 --alpha 0.05 --sides 2 --sd 10 --delta 5</code>
Rationale	Verifies the core superiority formula against the independently derived reference: $n/arm = \text{ceiling}(2 \cdot 10^2 \cdot (z_{0.975} + z_{0.80})^2 / 5^2) = 63$.
Pass criterion	n treatment = 63, n control = 63, n TOTAL = 126.

TC-CLIN-MEANS-003 Two-sided z-value reported

Tests UR(s)	UR-CLIN-003
Command / action	<code>jrc_clinical_ss_means --framework superiority --power 0.80 --alpha 0.05 --sides 2 --sd 10 --delta 5</code>
Rationale	The sidedness convention is the guide's #1 pitfall; the report must state it and the z it implies.
Pass criterion	Output contains '2-sided' and $z = 1.9600$.

TC-CLIN-MEANS-004 Non-inferiority numeric correctness

Tests UR(s)	UR-CLIN-004
Command / action	<code>jrc_clinical_ss_means --framework non_inferiority --power 0.90 --alpha 0.025 --sides 1 --sd 8 --margin 4</code>
Rationale	Verifies the NI branch: $n/arm = \text{ceiling}(2 \cdot 8^2 \cdot (z_{0.975} + z_{0.90})^2 / 4^2) = 85$ (alpha 0.025 one-sided).
Pass criterion	n treatment = 85, n control = 85, n TOTAL = 170.

TC-CLIN-MEANS-005 Dropout inflation

Tests UR(s)	UR-CLIN-005
Command / action	<code>jrc_clinical_ss_means --framework non_inferiority --power 0.90 --alpha 0.025 --sides 1 --sd 8 --margin 4 --dropout 0.15</code>
Rationale	Enrolled N is a script responsibility (never GUI arithmetic): $\text{ceiling}(85/0.85) = 100$ per arm.
Pass criterion	Output contains 'ENROLL 100 + 100 = 200'.

TC-CLIN-MEANS-006 Equivalence (TOST) numeric correctness

Tests UR(s)	UR-CLIN-006
Command / action	<code>jrc_clinical_ss_means --framework equivalence --power 0.80 --alpha 0.05 --sides 1 --sd 10 --margin 5</code>
Rationale	Verifies the TOST branch: $n/arm = \text{ceiling}(2 \cdot 10^2 \cdot (z_{0.95} + z_{0.90})^2 / 5^2) = 69$.
Pass criterion	n TOTAL = 138. Output contains 'equivalence (TOST)'.

TC-CLIN-MEANS-007 Allocation ratio 2:1

Tests UR(s)	UR-CLIN-007
Command / action	<code>jrc_clinical_ss_means --framework superiority --power 0.80 --alpha 0.05 --sides 2 --sd 10 --delta 5 --ratio 2</code>
Rationale	Verifies the $(1+1/k)$ allocation factor: control = $\text{ceiling}(1.5 \cdot 100 \cdot 2.801585^2 / 25) = 48$; treatment = 95.
Pass criterion	n treatment = 95, n control = 48.

TC-CLIN-MEANS-008 Sensitivity table scales with SD²

Tests UR(s)	UR-CLIN-008
Command / action	<code>jrc_clinical_ss_means --framework superiority --power 0.80 --alpha 0.05 --sides 2 --sd 10 --delta 5 --sensitivity</code>
Rationale	The SD is the most uncertain design input; the table must recompute n per scenario in the

	validated layer ($n \propto SD^2$).
Pass criterion	5 scenario rows; totals 82 at SD=8 and 182 at SD=12; assumed row marked.

TC-CLIN-MEANS-009 Superiority requires --delta

Tests UR(s)	UR-CLIN-009
Command / action	<code>jrc_clinical_ss_means --framework superiority --power 0.80 --alpha 0.05 --sides 2 --sd 10</code>
Rationale	A superiority design without delta is meaningless; the script must refuse rather than assume.
Pass criterion	Exit code $\neq 0$. 'requires --delta' in combined stdout/stderr.

TC-CLIN-MEANS-010 Non-inferiority requires --margin

Tests UR(s)	UR-CLIN-010
Command / action	<code>jrc_clinical_ss_means --framework non_inferiority --power 0.90 --alpha 0.025 --sides 1 --sd 8</code>
Rationale	The NI margin is the defining design input; it must be explicit.
Pass criterion	Exit code $\neq 0$. 'requires --margin' in combined stdout/stderr.

TC-CLIN-MEANS-011 Equivalence needs |delta| < margin

Tests UR(s)	UR-CLIN-011
Command / action	<code>jrc_clinical_ss_means --framework equivalence --power 0.80 --alpha 0.05 --sides 1 --sd 10 --margin 5 --delta 5</code>
Rationale	$ \text{delta} \geq \text{margin}$ makes the required n infinite; the script must refuse with a clear message.
Pass criterion	Exit code $\neq 0$. ' $ \text{delta} < \text{margin}$ ' in combined stdout/stderr.

TC-CLIN-MEANS-012 --power out of range

Tests UR(s)	UR-CLIN-012
Command / action	<code>jrc_clinical_ss_means --framework superiority --power 1.5 --alpha 0.05 --sides 2 --sd 10 --delta 5</code>
Rationale	Guards against a mistyped power silently producing nonsense.
Pass criterion	Exit code $\neq 0$. '--power' in combined stdout/stderr.

TC-CLIN-MEANS-013 Unknown flag rejected

Tests UR(s)	UR-CLIN-013
Command / action	<code>jrc_clinical_ss_means --framework superiority --power 0.80 --alpha 0.05 --sides 2 --sd 10 --delta 5 --bogus 1</code>
Rationale	A typo in a flag name must never be silently ignored in a validated tool.
Pass criterion	Exit code $\neq 0$. 'Unknown argument' in combined stdout/stderr.

TC-CLIN-MEANS-014 Bypass protection

Tests UR(s)	UR-CLIN-014
Command / action	<code>Rscript repos/clinical/R/jrc_clinical_ss_means.R ...</code> (RENV_PATHS_ROOT unset)
Rationale	Confirms that running the R script outside the validated jrrun path stops with the guardrail message.
Pass criterion	Exit code $\neq 0$. 'RENV_PATHS_ROOT' in combined stdout/stderr.

Test Cases — jrc_clinical_ss_props

TC-CLIN-PROPS-001 Superiority happy path — exit 0

Tests UR(s)	UR-CLIN-015
Command / action	<code>jrc_clinical_ss_props --framework superiority --power 0.80 --alpha 0.05 --sides 2 --p-control 0.70 --p-treat 0.85</code>
Rationale	Confirms the script accepts standard valid inputs and completes successfully.
Pass criterion	Exit code = 0. Output contains 'Clinical sample size' and 'binary endpoint'.

TC-CLIN-PROPS-002 Superiority numeric correctness

Tests UR(s)	UR-CLIN-016
Command / action	<code>jrc_clinical_ss_props --framework superiority --power 0.80 --alpha 0.05 --sides 2 --p-control 0.70 --p-treat 0.85</code>
Rationale	Verifies the core formula: $V = 0.85 \cdot 0.15 + 0.70 \cdot 0.30 = 0.3375$; $n/arm = \text{ceiling}(0.3375 \cdot (z0.975 + z0.80)^2 / 0.15^2) = 118$.
Pass criterion	$n \text{ treatment} = 118$, $n \text{ control} = 118$, $n \text{ TOTAL} = 236$.

TC-CLIN-PROPS-003 Non-inferiority numeric, default p-treat

Tests UR(s)	UR-CLIN-017
Command / action	<code>jrc_clinical_ss_props --framework non_inferiority --power 0.80 --alpha 0.025 --sides 1 --p-control 0.85 --margin 0.10</code>
Rationale	Verifies the NI branch and the documented default: $V = 2 \cdot 0.85 \cdot 0.15 = 0.255$; $n/arm = \text{ceiling}(0.255 \cdot 7.848886 / 0.01) = 201$.
Pass criterion	$n/arm = 201$, $n \text{ TOTAL} = 402$; 'assumed equal to control' in output.

TC-CLIN-PROPS-004 Equivalence (TOST) numeric correctness

Tests UR(s)	UR-CLIN-018
Command / action	<code>jrc_clinical_ss_props --framework equivalence --power 0.80 --alpha 0.05 --sides 1 --p-control 0.80 --margin 0.15</code>
Rationale	Verifies the TOST branch: $V = 2 \cdot 0.80 \cdot 0.20 = 0.32$; $n/arm = \text{ceiling}(0.32 \cdot (z0.95 + z0.90)^2 / 0.15^2) = 122$.
Pass criterion	$n \text{ TOTAL} = 244$. Output contains 'equivalence (TOST)'.

TC-CLIN-PROPS-005 Superiority requires --p-treat

Tests UR(s)	UR-CLIN-019
Command / action	<code>jrc_clinical_ss_props --framework superiority --power 0.80 --alpha 0.05 --sides 2 --p-control 0.70</code>
Rationale	A superiority design needs the treatment rate; the NI/equivalence default must not leak into superiority.
Pass criterion	Exit code $\neq 0$. 'requires --p-treat' in combined stdout/stderr.

TC-CLIN-PROPS-006 Equal rates rejected for superiority

Tests UR(s)	UR-CLIN-020
Command / action	<code>jrc_clinical_ss_props --framework superiority --power 0.80 --alpha 0.05 --sides 2 --p-control 0.70 --p-treat 0.70</code>
Rationale	Zero effect size gives an infinite n; the script must refuse.
Pass criterion	Exit code $\neq 0$. 'different from p-control' in combined stdout/stderr.

TC-CLIN-PROPS-007 Non-inferiority requires --margin

Tests UR(s)	UR-CLIN-021
Command / action	<code>jrc_clinical_ss_props --framework non_inferiority --power 0.80 --alpha 0.025 --sides 1 --p-control 0.85</code>
Rationale	The NI margin is the defining design input; it must be explicit.
Pass criterion	Exit code $\neq 0$. 'requires --margin' in combined stdout/stderr.

TC-CLIN-PROPS-008 --p-control out of range

Tests UR(s)	UR-CLIN-022
Command / action	<code>jrc_clinical_ss_props --framework non_inferiority --power 0.80 --</code>

	alpha 0.025 --sides 1 --p-control 1.2 --margin 0.10
Rationale	Guards against an impossible rate silently producing nonsense.
Pass criterion	Exit code $\neq$ 0. '--p-control' in combined stdout/stderr.

TC-CLIN-PROPS-009 Sensitivity table over the control rate

Tests UR(s)	UR-CLIN-023
Command / action	jrc_clinical_ss_props --framework equivalence --power 0.80 --alpha 0.05 --sides 1 --p-control 0.80 --margin 0.15 --sensitivity
Rationale	The control rate is the most uncertain design input; scenarios must be recomputed in the validated layer.
Pass criterion	5 scenario rows (0.64...0.96); assumed row marked.

TC-CLIN-PROPS-010 Dropout inflation

Tests UR(s)	UR-CLIN-024
Command / action	jrc_clinical_ss_props --framework non_inferiority --power 0.80 --alpha 0.025 --sides 1 --p-control 0.85 --margin 0.10 --dropout 0.10
Rationale	Enrolled N is a script responsibility: $\text{ceiling}(201/0.90) = 224$ per arm.
Pass criterion	Output contains 'ENROLL 224 + 224 = 448'.

TC-CLIN-PROPS-011 Bypass protection

Tests UR(s)	UR-CLIN-025
Command / action	Rscript repos/clinical/R/jrc_clinical_ss_props.R ... (RENV_PATHS_ROOT unset)
Rationale	Confirms that running the R script outside the validated jrrun path stops with the guardrail message.
Pass criterion	Exit code $\neq$ 0. 'RENV_PATHS_ROOT' in combined stdout/stderr.

Test Cases — jrc_clinical_ss_survival

TC-CLIN-SURV-001 Superiority happy path — exit 0

Tests UR(s)	UR-CLIN-026
Command / action	<code>jrc_clinical_ss_survival --framework superiority --power 0.80 --alpha 0.05 --sides 2 --hr 0.70 --event-prob 0.6</code>
Rationale	Confirms the script accepts standard valid inputs and completes successfully.
Pass criterion	Exit code = 0. Output contains 'Clinical sample size' and 'time-to-event endpoint'.

TC-CLIN-SURV-002 Superiority numeric correctness

Tests UR(s)	UR-CLIN-027
Command / action	<code>jrc_clinical_ss_survival --framework superiority --power 0.80 --alpha 0.05 --sides 2 --hr 0.70 --event-prob 0.6</code>
Rationale	Verifies the events formula: $E = \text{ceiling}(4 \cdot (z0.975 + z0.80)^2 / \log(0.70)^2) = 247$; $n/\text{arm} = \text{ceiling}((246.86/0.6)/2) = 206$.
Pass criterion	Events required = 247, n treatment = 206, n control = 206, n TOTAL = 412.

TC-CLIN-SURV-003 Non-inferiority numeric correctness

Tests UR(s)	UR-CLIN-028
Command / action	<code>jrc_clinical_ss_survival --framework non_inferiority --power 0.80 --alpha 0.025 --sides 1 --hr 1.0 --margin 1.3 --event-prob 0.5</code>
Rationale	Verifies the NI branch: $E = \text{ceiling}(4 \cdot 7.848886 / \log(1.3)^2) = 457$; $n/\text{arm} = \text{ceiling}((456.10/0.5)/2) = 457$.
Pass criterion	Events required = 457, n TOTAL = 914.

TC-CLIN-SURV-004 Equivalence not offered

Tests UR(s)	UR-CLIN-029
Command / action	<code>jrc_clinical_ss_survival --framework equivalence --power 0.80 --alpha 0.05 --sides 1 --hr 1.0 --event-prob 0.5</code>
Rationale	Equivalence is deliberately out of scope for time-to-event endpoints in this module.
Pass criterion	Exit code $\neq 0$. 'not offered' in combined stdout/stderr.

TC-CLIN-SURV-005 Superiority with HR = 1 rejected

Tests UR(s)	UR-CLIN-030
Command / action	<code>jrc_clinical_ss_survival --framework superiority --power 0.80 --alpha 0.05 --sides 2 --hr 1 --event-prob 0.6</code>
Rationale	HR = 1 means no effect; required events would be infinite.
Pass criterion	Exit code $\neq 0$. 'different from 1' in combined stdout/stderr.

TC-CLIN-SURV-006 NI margin must exceed 1

Tests UR(s)	UR-CLIN-031
Command / action	<code>jrc_clinical_ss_survival --framework non_inferiority --power 0.80 --alpha 0.025 --sides 1 --hr 1.0 --margin 0.9 --event-prob 0.5</code>
Rationale	An NI margin ≤ 1 is not a non-inferiority margin on a hazard ratio.
Pass criterion	Exit code $\neq 0$. '--margin' in combined stdout/stderr.

TC-CLIN-SURV-007 NI needs true HR inside the margin

Tests UR(s)	UR-CLIN-032
Command / action	<code>jrc_clinical_ss_survival --framework non_inferiority --power 0.80 --alpha 0.025 --sides 1 --hr 1.4 --margin 1.3 --event-prob 0.5</code>
Rationale	A true HR at or beyond the margin cannot demonstrate non-inferiority at any n.
Pass criterion	Exit code $\neq 0$. '--hr < --margin' in combined stdout/stderr.

TC-CLIN-SURV-008 --event-prob required

Tests UR(s)	UR-CLIN-033
Command / action	<code>jrc_clinical_ss_survival --framework superiority --power 0.80 --alpha 0.05 --sides 2 --hr 0.70</code>

Rationale	Events cannot be converted to subjects without the event probability.
Pass criterion	Exit code $\neq$ 0. '--event-prob' in combined stdout/stderr.

TC-CLIN-SURV-009 Sensitivity table over the event probability

Tests UR(s)	UR-CLIN-034
Command / action	<code>jrc_clinical_ss_survival --framework superiority --power 0.80 --alpha 0.05 --sides 2 --hr 0.70 --event-prob 0.6 --sensitivity</code>
Rationale	The event probability is the most uncertain design input; scenarios must be recomputed in the validated layer.
Pass criterion	5 scenario rows (0.48...0.72); assumed row marked.

TC-CLIN-SURV-010 Dropout inflation

Tests UR(s)	UR-CLIN-035
Command / action	<code>jrc_clinical_ss_survival --framework superiority --power 0.80 --alpha 0.05 --sides 2 --hr 0.70 --event-prob 0.6 --dropout 0.20</code>
Rationale	Enrolled N is a script responsibility: $\text{ceiling}(206/0.80) = 258$ per arm.
Pass criterion	Output contains 'ENROLL 258 + 258 = 516'.

TC-CLIN-SURV-011 Bypass protection

Tests UR(s)	UR-CLIN-036
Command / action	<code>Rscript repos/clinical/R/jrc_clinical_ss_survival.R ...</code> (RENV_PATHS_ROOT unset)
Rationale	Confirms that running the R script outside the validated jrrun path stops with the guardrail message.
Pass criterion	Exit code $\neq$ 0. 'RENV_PATHS_ROOT' in combined stdout/stderr.

Test Cases — jrc_clinical_dx_accuracy

TC-CLIN-DXACC-001 Valid inputs — exit 0

Tests UR(s)	UR-CLIN-037
Command / action	jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv
Rationale	Confirms the script accepts a standard valid dataset and completes.
Pass criterion	Exit code = 0. Output contains 'Diagnostic accuracy' and 'reference standard'.

TC-CLIN-DXACC-002 2x2 contingency table

Tests UR(s)	UR-CLIN-038
Command / action	jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv
Rationale	The 2x2 is the basis of every downstream measure; an error here propagates to all of them.
Pass criterion	TP 79, FP 6 / FN 16, TN 99; margins 95 reference-positive, 105 reference-negative, 200 total.

TC-CLIN-DXACC-003 Sensitivity with exact interval

Tests UR(s)	UR-CLIN-039
Command / action	jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv
Rationale	Sensitivity is the primary measure FDA (2007) requires with a CI.
Pass criterion	Sensitivity = 0.8316 (79/95), 95% CI (0.7410, 0.9006).

TC-CLIN-DXACC-004 Specificity with exact interval

Tests UR(s)	UR-CLIN-040
Command / action	jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv
Rationale	Specificity is the co-primary measure FDA (2007) requires with a CI.
Pass criterion	Specificity = 0.9429 (99/105), 95% CI (0.8798, 0.9787).

TC-CLIN-DXACC-005 Overall accuracy with exact interval

Tests UR(s)	UR-CLIN-041
Command / action	jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv
Rationale	Accuracy is reported for completeness; the report also carries the FDA caution against relying on it alone.
Pass criterion	Accuracy = 0.8900 (178/200), 95% CI (0.8382, 0.9298).

TC-CLIN-DXACC-006 Positive likelihood ratio with log interval

Tests UR(s)	UR-CLIN-042
Command / action	jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv
Rationale	Likelihood ratios need an interval on the log scale; a Wald interval on the ratio itself can cover negative values.
Pass criterion	LR+ = 14.5526, 95% CI (6.6563, 31.8163).

TC-CLIN-DXACC-007 Negative likelihood ratio with log interval

Tests UR(s)	UR-CLIN-043
Command / action	jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv
Rationale	As TC-CLIN-DXACC-006, for the negative likelihood ratio.
Pass criterion	LR- = 0.1786, 95% CI (0.1140, 0.2799).

TC-CLIN-DXACC-008 PPV/NPV at the study prevalence, with caveat

Tests UR(s)	UR-CLIN-044
Command / action	jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv
Rationale	PPV/NPV read straight off a 2x2 are the most commonly misreported quantities in diagnostic submissions; the caveat must be unmissable.
Pass criterion	PPV = 0.9294 (79/85), NPV = 0.8609 (99/115); output states 'study prevalence (0.4750)' and 'valid ONLY if'.

TC-CLIN-DXACC-009 Wilson interval option

Tests UR(s)	UR-CLIN-045
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Command / action	<code>jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv --ci wilson</code>
Rationale	Wilson is the shorter interval with better mean coverage; it must sit strictly inside the exact interval it replaces.
Pass criterion	Sensitivity estimate unchanged at 0.8316; 95% CI (0.7438, 0.8936), strictly inside the exact (0.7410, 0.9006).

TC-CLIN-DXACC-010 Bayes-adjusted PPV/NPV at a stated prevalence

Tests UR(s)	UR-CLIN-046
Command / action	<code>jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv --prevalence 0.02</code>
Rationale	This is the correct reporting for a case-control or enriched design, per FDA (2007). It is also the clearest demonstration of why the unadjusted values mislead: the same test reads PPV 0.93 at the study's 47.5% prevalence and 0.23 at 2%.
Pass criterion	PPV (adj) = 0.2290, 95% CI (0.1117, 0.4637); NPV (adj) = 0.9964. The unadjusted PPV 0.9294 is still shown.

TC-CLIN-DXACC-011 Confidence level honoured

Tests UR(s)	UR-CLIN-047
Command / action	<code>jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv --conf 0.90</code>
Rationale	Confirms the confidence level reaches the interval maths rather than only the printed label.
Pass criterion	Point estimate unchanged; both 90% bounds strictly inside the corresponding 95% bounds.

TC-CLIN-DXACC-012 Non-standard labels via --positive

Tests UR(s)	UR-CLIN-048
Command / action	<code>jrc_clinical_dx_accuracy dx_custom_labels.csv --positive detected</code>
Rationale	Real assay exports rarely use 1/0; the escape hatch must work.
Pass criterion	Exit code = 0. Sensitivity = 0.8000 (4/5), specificity = 0.8000 (4/5).

TC-CLIN-DXACC-013 Rows with missing data dropped and reported

Tests UR(s)	UR-CLIN-049
Command / action	<code>jrc_clinical_dx_accuracy dx_missing_rows.csv</code>
Rationale	Silently dropping subjects would misstate the denominator; the count must be visible in the report.
Pass criterion	Exit code = 0. Output contains '4 evaluable' and '2 row(s) dropped'.

TC-CLIN-DXACC-014 Invalid --ci rejected

Tests UR(s)	UR-CLIN-050
Command / action	<code>jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv --ci bogus</code>
Rationale	Guards against a silent fallback to an unintended interval method.
Pass criterion	Exit code != 0. Output contains '--ci must be'.

TC-CLIN-DXACC-015 Out-of-range --prevalence rejected

Tests UR(s)	UR-CLIN-051
Command / action	<code>jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv --prevalence 1.5</code>
Rationale	A prevalence outside (0,1) is meaningless and would silently produce nonsense Bayes values.
Pass criterion	Exit code != 0. Output names '--prevalence'.

TC-CLIN-DXACC-016 Missing required column rejected

Tests UR(s)	UR-CLIN-052
Command / action	<code>jrc_clinical_dx_accuracy dx_roc_n200_seed7.csv</code>
Rationale	The ROC fixture carries 'score' rather than 'result'; the error must name what is missing rather than fail obscurely.
Pass criterion	Exit code != 0. Output contains 'Missing column(s): result'.

TC-CLIN-DXACC-017 Duplicate subject id rejected

Tests UR(s)	UR-CLIN-053
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Command / action	<code>jrc_clinical_dx_accuracy dx_dup_id.csv</code>
Rationale	A duplicated subject would be double-counted in the 2x2, inflating the apparent sample size.
Pass criterion	Exit code != 0. Output contains 'Duplicate id(s) found'.

TC-CLIN-DXACC-018 Unrecognised labels rejected

Tests UR(s)	UR-CLIN-054
Command / action	<code>jrc_clinical_dx_accuracy dx_bad_labels.csv</code>
Rationale	Guessing which of two unknown labels means 'positive' could silently invert sensitivity and specificity.
Pass criterion	Exit code != 0. Output contains 'does not recognise'.

TC-CLIN-DXACC-019 Absent reference group rejected

Tests UR(s)	UR-CLIN-055
Command / action	<code>jrc_clinical_dx_accuracy dx_all_reference_positive.csv</code>
Rationale	Specificity is undefined without reference-negative subjects; reporting 0/0 would be worse than refusing.
Pass criterion	Exit code != 0. Output contains 'specificity is undefined'.

TC-CLIN-DXACC-020 Bypass guardrail

Tests UR(s)	UR-CLIN-056
Command / action	<code>Rscript --vanilla jrc_clinical_dx_accuracy.R (no RENV_PATHS_ROOT)</code>
Rationale	Confirms that running the R script outside the validated jrrun path stops with the guardrail message.
Pass criterion	Exit code != 0. 'RENV_PATHS_ROOT' in combined stdout/stderr.

Test Cases — jrc_clinical_dx_ss

TC-CLIN-DXSS-001 Precision method happy path — exit 0

Tests UR(s)	UR-CLIN-057
Command / action	jrc_clinical_dx_ss --method precision --sens-expected 0.90 --spec-expected 0.95 --halfwidth 0.05 --prevalence 0.10
Rationale	Confirms the script accepts standard valid inputs and completes.
Pass criterion	Exit code = 0. Output contains 'Clinical sample size' and 'diagnostic accuracy'.

TC-CLIN-DXSS-002 Precision numeric correctness (Buderer 1996)

Tests UR(s)	UR-CLIN-058
Command / action	jrc_clinical_dx_ss --method precision --sens-expected 0.90 --spec-expected 0.95 --halfwidth 0.05 --prevalence 0.10
Rationale	Reproduces the published worked example: $n_{pos} = z^2 * 0.9 * 0.1 / 0.05^2 = 138.29 \rightarrow 139$; $n_{neg} = z^2 * 0.95 * 0.05 / 0.05^2 = 72.99 \rightarrow 73$; $N = 138.29 / 0.10 = 1382.93 \rightarrow 1383$.
Pass criterion	n reference + = 139, n reference - = 73, N TOTAL = 1383.

TC-CLIN-DXSS-003 Hypothesis numeric correctness

Tests UR(s)	UR-CLIN-059
Command / action	jrc_clinical_dx_ss --method hypothesis --sens-expected 0.90 --spec-expected 0.95 --sens-goal 0.80 --spec-goal 0.90 --power 0.80 --alpha 0.025 --sides 1 --prevalence 0.10
Rationale	$n_{pos} = (z_a * \sqrt{.8 * .2} + z_b * \sqrt{.9 * .1})^2 / (.9 - .8)^2 = 107.43 \rightarrow 108$; $n_{neg} = (z_a * \sqrt{.9 * .1} + z_b * \sqrt{.95 * .05})^2 / (.95 - .90)^2 = 238.02 \rightarrow 239$; $N = \max(107.43 / .10, 238.02 / .90) \rightarrow 1075$.
Pass criterion	n reference + = 108, n reference - = 239, N TOTAL = 1075.

TC-CLIN-DXSS-004 Binding arm identified

Tests UR(s)	UR-CLIN-060
Command / action	jrc_clinical_dx_ss --method precision --sens-expected 0.90 --spec-expected 0.95 --halfwidth 0.05 --prevalence 0.10
Rationale	At low prevalence the reference-positive arm binds; naming it tells the designer which assumption to attack to shrink the study.
Pass criterion	Output contains 'Binding arm : sensitivity (reference-positive)'.

TC-CLIN-DXSS-005 Rarer prevalence raises enrolment, not arm size

Tests UR(s)	UR-CLIN-061
Command / action	jrc_clinical_dx_ss --method precision --sens-expected 0.90 --spec-expected 0.95 --halfwidth 0.05 --prevalence 0.05
Rationale	Separates the two effects: arm requirements depend only on the target precision; only the enrolment needed to reach them depends on prevalence. Halving prevalence doubles enrolment.
Pass criterion	n reference + = 139 (unchanged); N TOTAL = 2766, greater than the 1383 of TC-CLIN-DXSS-002.

TC-CLIN-DXSS-006 Prevalence sensitivity table

Tests UR(s)	UR-CLIN-062
Command / action	jrc_clinical_dx_ss --method precision --sens-expected 0.90 --spec-expected 0.95 --halfwidth 0.05 --prevalence 0.10 --sensitivity
Rationale	Prevalence is the assumption most often wrong and most punishing when it is; the table makes the exposure visible at design time.
Pass criterion	Exit code = 0. Five rows (prevalence x 0.5, 0.75, 1.0, 1.5, 2.0); the assumed row marked '<- assumed'.

TC-CLIN-DXSS-007 Dropout inflation

Tests UR(s)	UR-CLIN-063
Command / action	jrc_clinical_dx_ss --method hypothesis --sens-expected 0.90 --spec-expected 0.95 --sens-goal 0.80 --spec-goal 0.90 --power 0.80 --alpha 0.025 --sides 1 --prevalence 0.10 --dropout 0.10
Rationale	enrolled = ceiling(1075 / 0.90) = 1195.

Pass criterion	Output contains 'ENROLL 1195 subjects'.
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TC-CLIN-DXSS-008 Sidedness default differs by method

Tests UR(s)	UR-CLIN-064
Command / action	<code>jrc_clinical_dx_ss --method precision --sens-expected 0.90 --spec-expected 0.95 --halfwidth 0.05 --prevalence 0.10 / jrc_clinical_dx_ss --method hypothesis --sens-expected 0.90 --spec-expected 0.95 --sens-goal 0.80 --spec-goal 0.90 --power 0.80 --prevalence 0.10</code>
Rationale	A CI half-width is inherently two-sided; a performance-goal test is conventionally one-sided. Defaulting wrongly would silently mis-size the study.
Pass criterion	Precision output contains 'two-sided, z = 1.9600'; hypothesis output without explicit --sides contains '1-sided'.

TC-CLIN-DXSS-009 Precision requires --halfwidth

Tests UR(s)	UR-CLIN-065
Command / action	<code>jrc_clinical_dx_ss --method precision --sens-expected 0.90 --spec-expected 0.95 --prevalence 0.10</code>
Rationale	There is no defensible default target precision; it must be stated.
Pass criterion	Exit code != 0. Output contains 'requires --halfwidth'.

TC-CLIN-DXSS-010 Hypothesis requires --sens-goal

Tests UR(s)	UR-CLIN-066
Command / action	<code>jrc_clinical_dx_ss --method hypothesis --sens-expected 0.90 --spec-expected 0.95 --spec-goal 0.90 --prevalence 0.10</code>
Rationale	There is no defensible default performance goal; it must be stated.
Pass criterion	Exit code != 0. Output contains 'requires --sens-goal'.

TC-CLIN-DXSS-011 Goal must be below the expected value

Tests UR(s)	UR-CLIN-067
Command / action	<code>jrc_clinical_dx_ss --method hypothesis --sens-expected 0.90 --spec-expected 0.95 --sens-goal 0.95 --spec-goal 0.90 --prevalence 0.10</code>
Rationale	A goal at or above the expected value cannot be powered against; the formula would divide by a non-positive difference.
Pass criterion	Exit code != 0. Output contains '--sens-goal must be strictly less than'.

TC-CLIN-DXSS-012 Out-of-range --prevalence rejected

Tests UR(s)	UR-CLIN-068
Command / action	<code>jrc_clinical_dx_ss --method precision --sens-expected 0.90 --spec-expected 0.95 --halfwidth 0.05 --prevalence 1.2</code>
Rationale	A prevalence outside (0,1) is meaningless and would produce a negative or infinite enrolment.
Pass criterion	Exit code != 0. Output names '--prevalence'.

TC-CLIN-DXSS-013 Invalid --method rejected

Tests UR(s)	UR-CLIN-069
Command / action	<code>jrc_clinical_dx_ss --method bogus</code>
Rationale	Guards against a silent fallback to an unintended sizing method.
Pass criterion	Exit code != 0. Output contains '--method must be one of'.

TC-CLIN-DXSS-014 Bypass guardrail

Tests UR(s)	UR-CLIN-070
Command / action	<code>Rscript --vanilla jrc_clinical_dx_ss.R (no RENV_PATHS_ROOT)</code>
Rationale	Confirms that running the R script outside the validated jrrun path stops with the guardrail message.
Pass criterion	Exit code != 0. 'RENV_PATHS_ROOT' in combined stdout/stderr.

Test Cases — jrc_clinical_dx_roc

TC-CLIN-DXROC-001 Valid inputs — exit 0

Tests UR(s)	UR-CLIN-071
Command / action	jrc_clinical_dx_roc dx_roc_n200_seed7.csv
Rationale	Confirms the script accepts a standard valid dataset and completes.
Pass criterion	Exit code = 0. Output contains 'ROC analysis'; 80 reference-positive and 120 reference-negative subjects reported.

TC-CLIN-DXROC-002 ROC plot written to ~/Downloads/

Tests UR(s)	UR-CLIN-072
Command / action	jrc_clinical_dx_roc dx_roc_n200_seed7.csv
Rationale	The plot is a deliverable of the script, not a side effect.
Pass criterion	A file matching *_jrc_clinical_dx_roc.png is created in ~/Downloads/ during the run.

TC-CLIN-DXROC-003 PUBLISHED ANCHOR — Hanley & McNeil (1982) AUC

Tests UR(s)	UR-CLIN-073
Command / action	jrc_clinical_dx_roc dx_roc_hanley_mcnail_1982.csv
Rationale	The AUC and its DeLong variance are implemented in base R in this project rather than delegated to a package, so the acceptance value is taken from the ORIGINAL LITERATURE rather than from our own output. Hanley JA, McNeil BJ (1982), Radiology 143:29-36, Table 1: 51 abnormal and 58 normal cases on a 5-point diagnostic confidence rating; the paper reports an area of 0.893. Note the paper's SE is deliberately NOT used as an acceptance criterion: it is a binormal-based estimator, whereas this script reports the nonparametric DeLong SE.
Pass criterion	51 reference-positive, 58 reference-negative. AUC = 0.8932, which equals the published 0.893 at the paper's three decimal places.

TC-CLIN-DXROC-004 AUC, DeLong SE and confidence interval

Tests UR(s)	UR-CLIN-074
Command / action	jrc_clinical_dx_roc dx_roc_n200_seed7.csv
Rationale	The DeLong interval is the conventional nonparametric interval for an empirical AUC.
Pass criterion	AUC = 0.8268, SE = 0.0294, 95% CI (0.7691, 0.8845), labelled 'DeLong (raw scale)'.

TC-CLIN-DXROC-005 Tie-aware AUC kernel

Tests UR(s)	UR-CLIN-075
Command / action	jrc_clinical_dx_roc dx_roc_ties_n26_seed11.csv
Rationale	Integer scores 1-5 across 26 subjects make almost every comparison a tie. The expected value is obtainable only with the psi = 0.5 kernel; a kernel that ignored ties would not return it. This is what makes the empirical AUC equal the trapezoidal area under the ROC curve.
Pass criterion	12 reference-positive, 14 reference-negative. AUC = 0.5952.

TC-CLIN-DXROC-006 Youden-optimal operating point

Tests UR(s)	UR-CLIN-076
Command / action	jrc_clinical_dx_roc dx_roc_n200_seed7.csv
Rationale	Youden's J is the conventional summary of the best single cutoff.
Pass criterion	J = 0.5125, sensitivity = 0.5875, specificity = 0.9250.

TC-CLIN-DXROC-007 Cutoff reported as an observed value with a rule

Tests UR(s)	UR-CLIN-077
Command / action	jrc_clinical_dx_roc dx_roc_n200_seed7.csv
Rationale	An observed value is directly reportable and cannot fall outside the measured range. Other tools print the midpoint between adjacent observed values; both name the identical classifier, but only an explicit rule makes the report unambiguous.
Pass criterion	Output contains 'Cutoff : score >= 2.192 => positive'.

TC-CLIN-DXROC-008 Test of AUC = 0.5

Tests UR(s)	UR-CLIN-078
Command / action	<code>jrc_clinical_dx_roc dx_roc_n200_seed7.csv</code>
Rationale	AUC = 0.5 is the no-discrimination null; the test is the basic question asked of any new marker.
Pass criterion	Output contains 'H0: AUC = 0.5' and 'z = 11.1017'.

TC-CLIN-DXROC-009 Direction reverses the AUC about 0.5

Tests UR(s)	UR-CLIN-079
Command / action	<code>jrc_clinical_dx_roc dx_roc_n200_seed7.csv --direction lower</code>
Rationale	Reversing the assumed assay orientation must map AUC to 1 - AUC exactly; any other result indicates the orientation is not applied consistently.
Pass criterion	AUC = 0.1732 against 0.8268 for the default direction; the two sum to 1 within 1e-9.

TC-CLIN-DXROC-010 Degenerate cutoff reported honestly

Tests UR(s)	UR-CLIN-080
Command / action	<code>jrc_clinical_dx_roc dx_roc_n200_seed7.csv --direction lower</code>
Rationale	With the direction reversed, every candidate cutoff has $J \leq 0$, so the Youden optimum falls on an infinite sentinel threshold. Printing 'score $\leq \text{Inf} \Rightarrow$ positive' would be meaningless; the honest report is that there is no usable cutoff, plus a warning that the assay orientation is probably inverted.
Pass criterion	Exit code = 0. Output contains 'Cutoff : none' and 'AUC < 0.5', and suggests '--direction higher'. No infinite threshold is printed in the Youden section.

TC-CLIN-DXROC-011 Logit interval option

Tests UR(s)	UR-CLIN-081
Command / action	<code>jrc_clinical_dx_roc dx_roc_n200_seed7.csv --ci-method logit</code>
Rationale	The logit interval stays inside (0,1) and behaves better when the AUC approaches 1, where the raw-scale interval can overshoot.
Pass criterion	AUC unchanged at 0.8268; 95% CI (0.7614, 0.8772), labelled 'DeLong (logit-transformed)'.

TC-CLIN-DXROC-012 Confidence level honoured

Tests UR(s)	UR-CLIN-082
Command / action	<code>jrc_clinical_dx_roc dx_roc_n200_seed7.csv --conf 0.90</code>
Rationale	Confirms the confidence level reaches the interval maths rather than only the printed label.
Pass criterion	AUC unchanged at 0.8268; 90% CI (0.7784, 0.8752), strictly inside the 95% CI (0.7691, 0.8845).

TC-CLIN-DXROC-013 Invalid --direction rejected

Tests UR(s)	UR-CLIN-083
Command / action	<code>jrc_clinical_dx_roc dx_roc_n200_seed7.csv --direction sideways</code>
Rationale	Guards against a silent fallback to an unintended orientation, which would invert the entire analysis.
Pass criterion	Exit code $\neq 0$. Output contains '--direction must be'.

TC-CLIN-DXROC-014 Missing required column rejected

Tests UR(s)	UR-CLIN-084
Command / action	<code>jrc_clinical_dx_roc dx_accuracy_n200_seed42.csv</code>
Rationale	The accuracy fixture carries 'result' rather than 'score'; the error must name what is missing rather than fail obscurely.
Pass criterion	Exit code $\neq 0$. Output contains 'Missing column(s): score'.

TC-CLIN-DXROC-015 Duplicate subject id rejected

Tests UR(s)	UR-CLIN-085
Command / action	<code>jrc_clinical_dx_roc dx_dup_id.csv</code>
Rationale	A duplicated subject would be counted twice in the rank statistics, biasing the AUC and understating its variance.
Pass criterion	Exit code $\neq 0$. Output contains 'Duplicate id(s) found'.

TC-CLIN-DXROC-016 Absent reference group rejected

Tests UR(s)	UR-CLIN-086
Command / action	<code>jrc_clinical_dx_roc dx_all_reference_positive.csv</code>
Rationale	An ROC curve is undefined without both reference groups; there is nothing to discriminate between.
Pass criterion	Exit code != 0. Output contains 'At least 2 reference-negative'.

TC-CLIN-DXROC-017 Bypass guardrail

Tests UR(s)	UR-CLIN-087
Command / action	<code>Rscript --vanilla jrc_clinical_dx_roc.R (no RENV_PATHS_ROOT)</code>
Rationale	Confirms that running the R script outside the validated jrun path stops with the guardrail message.
Pass criterion	Exit code != 0. 'RENV_PATHS_ROOT' in combined stdout/stderr.

Requirements Traceability Matrix

Maps each test case to the user requirement(s) it covers.

Test Case ID	User Requirement(s)
TC-CLIN-MEANS-001	UR-CLIN-001
TC-CLIN-MEANS-002	UR-CLIN-002
TC-CLIN-MEANS-003	UR-CLIN-003
TC-CLIN-MEANS-004	UR-CLIN-004
TC-CLIN-MEANS-005	UR-CLIN-005
TC-CLIN-MEANS-006	UR-CLIN-006
TC-CLIN-MEANS-007	UR-CLIN-007
TC-CLIN-MEANS-008	UR-CLIN-008
TC-CLIN-MEANS-009	UR-CLIN-009
TC-CLIN-MEANS-010	UR-CLIN-010
TC-CLIN-MEANS-011	UR-CLIN-011
TC-CLIN-MEANS-012	UR-CLIN-012
TC-CLIN-MEANS-013	UR-CLIN-013
TC-CLIN-MEANS-014	UR-CLIN-014
TC-CLIN-PROPS-001	UR-CLIN-015
TC-CLIN-PROPS-002	UR-CLIN-016
TC-CLIN-PROPS-003	UR-CLIN-017
TC-CLIN-PROPS-004	UR-CLIN-018
TC-CLIN-PROPS-005	UR-CLIN-019
TC-CLIN-PROPS-006	UR-CLIN-020
TC-CLIN-PROPS-007	UR-CLIN-021
TC-CLIN-PROPS-008	UR-CLIN-022
TC-CLIN-PROPS-009	UR-CLIN-023
TC-CLIN-PROPS-010	UR-CLIN-024
TC-CLIN-PROPS-011	UR-CLIN-025
TC-CLIN-SURV-001	UR-CLIN-026
TC-CLIN-SURV-002	UR-CLIN-027
TC-CLIN-SURV-003	UR-CLIN-028
TC-CLIN-SURV-004	UR-CLIN-029
TC-CLIN-SURV-005	UR-CLIN-030
TC-CLIN-SURV-006	UR-CLIN-031
TC-CLIN-SURV-007	UR-CLIN-032
TC-CLIN-SURV-008	UR-CLIN-033
TC-CLIN-SURV-009	UR-CLIN-034
TC-CLIN-SURV-010	UR-CLIN-035
TC-CLIN-SURV-011	UR-CLIN-036
TC-CLIN-DXACC-001	UR-CLIN-037
TC-CLIN-DXACC-002	UR-CLIN-038
TC-CLIN-DXACC-003	UR-CLIN-039
TC-CLIN-DXACC-004	UR-CLIN-040
TC-CLIN-DXACC-005	UR-CLIN-041
TC-CLIN-DXACC-006	UR-CLIN-042
TC-CLIN-DXACC-007	UR-CLIN-043
TC-CLIN-DXACC-008	UR-CLIN-044
TC-CLIN-DXACC-009	UR-CLIN-045
TC-CLIN-DXACC-010	UR-CLIN-046
TC-CLIN-DXACC-011	UR-CLIN-047
TC-CLIN-DXACC-012	UR-CLIN-048
TC-CLIN-DXACC-013	UR-CLIN-049

TC-CLIN-DXACC-014	UR-CLIN-050
TC-CLIN-DXACC-015	UR-CLIN-051
TC-CLIN-DXACC-016	UR-CLIN-052
TC-CLIN-DXACC-017	UR-CLIN-053
TC-CLIN-DXACC-018	UR-CLIN-054
TC-CLIN-DXACC-019	UR-CLIN-055
TC-CLIN-DXACC-020	UR-CLIN-056
TC-CLIN-DXSS-001	UR-CLIN-057
TC-CLIN-DXSS-002	UR-CLIN-058
TC-CLIN-DXSS-003	UR-CLIN-059
TC-CLIN-DXSS-004	UR-CLIN-060
TC-CLIN-DXSS-005	UR-CLIN-061
TC-CLIN-DXSS-006	UR-CLIN-062
TC-CLIN-DXSS-007	UR-CLIN-063
TC-CLIN-DXSS-008	UR-CLIN-064
TC-CLIN-DXSS-009	UR-CLIN-065
TC-CLIN-DXSS-010	UR-CLIN-066
TC-CLIN-DXSS-011	UR-CLIN-067
TC-CLIN-DXSS-012	UR-CLIN-068
TC-CLIN-DXSS-013	UR-CLIN-069
TC-CLIN-DXSS-014	UR-CLIN-070
TC-CLIN-DXROC-001	UR-CLIN-071
TC-CLIN-DXROC-002	UR-CLIN-072
TC-CLIN-DXROC-003	UR-CLIN-073
TC-CLIN-DXROC-004	UR-CLIN-074
TC-CLIN-DXROC-005	UR-CLIN-075
TC-CLIN-DXROC-006	UR-CLIN-076
TC-CLIN-DXROC-007	UR-CLIN-077
TC-CLIN-DXROC-008	UR-CLIN-078
TC-CLIN-DXROC-009	UR-CLIN-079
TC-CLIN-DXROC-010	UR-CLIN-080
TC-CLIN-DXROC-011	UR-CLIN-081
TC-CLIN-DXROC-012	UR-CLIN-082
TC-CLIN-DXROC-013	UR-CLIN-083
TC-CLIN-DXROC-014	UR-CLIN-084
TC-CLIN-DXROC-015	UR-CLIN-085
TC-CLIN-DXROC-016	UR-CLIN-086
TC-CLIN-DXROC-017	UR-CLIN-087